

# Admire Theophilus Madyira

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## EXPERIENCE

### Software Engineer

Microsoft

August 2025 – Present

Redmond, WA

- Halved Edge Update latency and doubled bandwidth savings by cutting payload sizes from 400MB to 200MB via differential patches with hash-based fingerprints; shipped to 1B+ devices with end-to-end ownership in C++, Bash, Python, C#, and Kusto QL.
- Built AI agents integrated into Copilot and Edge release pipelines monitoring Kusto/ADX dashboards for build health, failure rates, and fleet telemetry, auto-triaging failures and cutting incident triage from 30+ min to ~5 min.
- Owned Linux incident response: resolved 10 incidents impacting ~20,000 devices in three months; first to ship YAML and Docker config validation into a nearly decade-old Smoke test pipeline.
- Designed code check-in policies validating ~400 PRs/week, blocking stale compliance checks and flagging API changes lacking privacy review; earned full pipeline ownership within 3 months.

### Software Engineer Intern

Microsoft

May 2024 – August 2024

Redmond, WA

- Designed and shipped a novel LLM council framework for model vulnerability detection, achieving ~90% accuracy on prompt-injection, data-leakage, and jailbreak exploits in Copilot; delivered one month ahead of deadline.
- Built an AI agent for bug-fixing and crash analysis, cutting engineer triage time by ~50% and securing a return offer.

### Software Engineer Intern

Microsoft

May 2023 – August 2023

Redmond, WA

- Optimized Microsoft Edge search results algorithm and improved latency in C++, contributing to a ~5% lift in user retention.
- Conducted an A/B experiment validating net positive revenue impact on Microsoft Bing.

### Machine Learning Engineer Intern

Columbia University

June 2022 – August 2022

New York, NY

- Trained RL motion-control policies (PPO, SAC, DQN) for autonomous vehicles in the CARLA simulator, improving trajectory stability and decision accuracy across diverse driving scenarios.

### Full Stack Engineer Apprentice

Google

June 2021 – May 2022

Remote

- Built a full-stack marketing website for a poultry business as a capstone in the 10-month Google Africa Developer Scholarship program, using React.js, Node.js, GraphQL, PostgreSQL, and REST APIs under Google engineer mentorship.

## SKILLS

**Languages:** C++, Python, C#, SQL, Java, TypeScript, JavaScript, Bash, PowerShell

**AI / ML:** AI Agents, LLM Fine-Tuning, RAG, NLP, Reinforcement Learning (PPO, SAC, DQN), MLOps, LLMops

**Systems & Infrastructure:** Distributed Systems, Ultra Low Latency Optimization, System Design, CI/CD, Infrastructure as Code, Observability, Incident Management, SRE

**Frameworks & Tools:** PyTorch, TensorFlow, Scikit-Learn, Kubernetes, Docker, Terraform, GitHub Actions, Kusto/ADX, Catch2

## PROJECTS

### QuantIQ

- Built a C++ trading system running 4 strategies (Breakout, MeanReversion, Momentum, SmaCrossover) against an Alpaca paper account across 1,255 bars; replays 5,020 bar-evaluations in 55.6 ms at -O3, with a multithreaded data feed, drawdown kill switch, and a published dashboard benchmarking each strategy against buy-and-hold.

### Agnostic LLM Watermarker

- Co-built a study comparing prompt-level and token-level LLM watermarking: prompt constraints against a logit-biased greenlist across ~6,000 GPT-3.5-Turbo generations; RoBERTa detectors recovered 86% of prompt-based vs. 66% of token-level watermarks.

## AWARDS & HONORS

- Rhodes Scholarship Finalist (2025): Selected as 1 of 10 national finalists for the post-graduate scholarship at Oxford University.
- 2x Walker Award in Mathematics & Statistics, Amherst College (2023, 2025).
- Presidential & National Scholarship, Gov. of Zimbabwe (2020): Full study-abroad funding after ranking 1st of 50,000+ students on national A-Level exams.

## EDUCATION

### Columbia University

MSc. in Computer Science.

Expected May 2028

### Amherst College

BA in Computer Science & Mathematics.

May 2025